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Cambridge IGCSE Combined and Co-ordinated Sciences

B4 Enzymes

Comprehensive Exam Preparation Guide

B4 Enzymes — Cambridge IGCSE Biology Study Guide

Chapter Overview

In this chapter you will:

- Learn about enzymes, and how they work as catalysts
- Investigate how temperature and pH affect the activity of enzymes
- Find out why enzymes are specific, and why temperature and pH affect them

All learners study some content in this chapter. In the syllabus, explaining enzyme action using the enzyme–substrate complex, explaining why enzymes are specific, and explaining (not just describing) the effects of temperature and pH are Supplement content. These parts are marked (Supplement) in this guide.

Important Terms

Term	Definition
Catalyst	A substance that increases the rate of a chemical reaction and is not changed by the reaction
Enzymes	Proteins that are involved in all metabolic reactions, where they function as biological catalysts
Biological catalyst	A catalyst made by living cells; enzymes are biological catalysts that speed up the reactions inside organisms
Metabolic reactions	The chemical reactions that take place all the time inside living organisms; almost every one is controlled by an enzyme
Substrate	The substance that an enzyme causes to react (for example, starch is the substrate of amylase)
Product	The new substance formed by a chemical reaction
Active site	The part of an enzyme molecule to which the substrate temporarily binds
Complementary	With a perfect mirror-image shape; the active site and its substrate have complementary shapes, so they fit together
Enzyme–substrate complex	The short-lived structure formed as the substrate binds temporarily to the active site of an enzyme
Specificity	Of enzymes, only able to act on a particular (specific) substrate
Lock and key model	The idea that a substrate fits an enzyme's active site like a key fits a lock; only a substrate with the complementary shape can fit
Carbohydrases	Enzymes that break down carbohydrates (for example, amylase, maltase, sucrase and lactase)
Protease	An enzyme that catalyses the breakdown of protein to amino acids
Lipases	Enzymes that break down lipids (fats and oils) into fatty acids and glycerol

Amylase	An enzyme that catalyses the breakdown of starch to maltose
Maltase	An enzyme that catalyses the breakdown of maltose to glucose
Sucrase	An enzyme that breaks down sucrose
Lactase	An enzyme that breaks down the milk sugar lactose into smaller sugar molecules that can be absorbed into the blood
Catalase	An enzyme that catalyses the breakdown of hydrogen peroxide to water and oxygen; it works inside all living cells
Hydrogen peroxide	A very dangerous substance made by many chemical reactions inside cells; catalase must break it down immediately
Lactose intolerance	Feeling ill (diarrhoea, abdominal pain, flatulence) after eating dairy foods, because the body does not make lactase and lactose is not broken down
Optimum	Best; for example, the optimum temperature of an enzyme is the temperature at which its activity is greatest
Optimum temperature	The temperature at which an enzyme works fastest; about 37 °C for enzymes from the human body
Optimum pH	The pH at which an enzyme works fastest; around pH 7 for most enzymes, but about pH 2 for a protease in the stomach
Denaturation	Damage to an enzyme's shape (including its active site) caused by high temperature or an unsuitable pH; the enzyme is denatured and can no longer bind its substrate or catalyse its reaction
Kinetic energy	Energy of movement; as temperature rises, enzyme and substrate molecules gain kinetic energy and move faster
Effective collision	A collision in which a substrate molecule enters an enzyme's active site, forms an enzyme–substrate complex and is converted to product
Range	The lowest to the highest value
Buffer solution	A liquid that has a known pH, and keeps that pH steady all the time, even if chemical reactions take place
Thymolphthalein	An indicator that is blue in alkaline conditions and loses its colour when the pH becomes less alkaline
Iodine solution	A brown (orange-brown) solution used to test for starch; it turns blue-black when starch is present
Anomalous result	A result that does not fit the pattern of the other results; it is left out when calculating a mean

B4.01 Biological Catalysts

Before You Start: What Happens in a Chemical Reaction?

Enzymes control chemical reactions, so the chapter begins by checking what you already know about reactions. The book gives five statements and says that two of them are wrong.

Q: Here are some statements about chemical reactions. Two of them are incorrect. Can you identify these incorrect statements? When you think you have identified them, change them so that they are correct. In a chemical reaction: one or more substances are changed to one or more new substances; a

large molecule may be split into smaller molecules; atoms of one element are rearranged to make atoms of other elements; small molecules may combine to form larger molecules; products are changed to reactants.

A: The two incorrect statements are:

- "Atoms of one element are rearranged to make atoms of other elements." This is wrong, because atoms are never changed into atoms of a different element in a chemical reaction. Corrected version: in a chemical reaction, atoms are rearranged to make new substances, but each atom stays as the same element.
- "Products are changed to reactants." This is the wrong way round. Corrected version: reactants are changed to products.

The other three statements are correct, and enzymes catalyse exactly these kinds of reactions: splitting large molecules into small ones (for example, starch into maltose) and joining small molecules into large ones (for example, amino acids into proteins).

Science in Context B4.01: The Dangers of Dairy Products

All mammals feed their young on milk (Figure B4.01 shows young guanacos feeding on milk from their mother). Milk is an excellent food. It contains carbohydrates, fats, proteins, minerals and vitamins.

Human babies are mammals too, so they feed on milk. To help them digest the carbohydrate in milk, they produce an enzyme called **lactase**. Lactase breaks a sugar called **lactose** into smaller molecules, which can then be absorbed into the blood.

Most people stop making lactase once they are old enough to eat food other than milk. Some people keep making lactase all their lives and can drink milk and eat dairy products with no problems. Everyone else is **intolerant to lactose**, and dairy foods make them feel ill:

1. They do not make lactase, so lactose is not broken down in the digestive system.
2. The lactose stays inside the alimentary canal (the tube that food passes through).
3. This makes water move into the alimentary canal by osmosis, and the extra water can cause diarrhoea.
4. Bacteria living in the alimentary canal feed on the lactose and produce gases as a waste product, which can cause abdominal pain and flatulence.

Solutions to lactose intolerance: the simplest solution is not to eat foods containing lactose. Another is to swallow a pill containing the enzyme lactase before eating dairy products, which may prevent the unpleasant symptoms.

Q: Discussion question 1 — If most people stop making lactase when they are adults, there must be some advantage in this. Discuss what this might be.

A: Possible ideas to discuss:

- Making an enzyme uses energy and raw materials (amino acids are needed to build the protein). An adult who no longer drinks milk does not need lactase, so stopping making it saves energy and amino acids for making other, more useful proteins.
- Long ago, before people kept animals for milk, adults had no milk to drink, so lactase was of no use after a child had stopped feeding on its mother's milk.
- Losing lactase may help a young mammal to stop feeding on its mother's milk (weaning), so the mother can save her energy and resources for her next baby.

What Is a Catalyst?

Many chemical reactions can be sped up by substances called **catalysts**. A catalyst alters the rate of a chemical reaction without being changed itself. Because it is not used up, a catalyst can be used again and again.

Enzymes as Biological Catalysts

Within any living organism, chemical reactions are taking place all the time. These are called **metabolic reactions**. Almost every metabolic reaction is controlled by catalysts called **enzymes**.

- Enzymes are **proteins** (so they are made of amino acids).
- Without enzymes, metabolic reactions would take place very slowly or not at all.
- Enzymes make sure that the rates of metabolic reactions are **fast enough to sustain life**.

Examples of Enzymes at Work

Digestion. Inside the intestines, large molecules are broken down to smaller ones in the process of digestion. These reactions are sped up by enzymes, and a different enzyme is needed for each kind of nutrient:

- starch → maltose (catalysed by amylase)
- protein → amino acids (catalysed by protease)

Plants. These enzymes are also found in plants. In germinating seeds, enzymes digest the food stores (mainly starch) to feed the growing seedling.

Catalase. Catalase works inside all the cells of living organisms. Many of the chemical reactions in cells produce hydrogen peroxide, which is a very dangerous substance and must be broken down immediately. Catalase breaks it down:

- hydrogen peroxide → water + oxygen (catalysed by catalase)

Building up molecules. Not all enzymes help to break things down. Many enzymes help to make large molecules from small ones. For example, enzymes help to link amino acids together to make proteins inside cells.

Naming Enzymes

Enzymes are named according to the reaction that they catalyse, and their names often end in **-ase**:

- Enzymes that break down **carbohydrates** are **carbohydrases**.
- Enzymes that break down **proteins** are **proteases**.
- Enzymes that break down **fats and oils (lipids)** are **lipases**.

Some enzymes have more specific names. The carbohydrase that breaks down **starch** is **amylase**, the one that breaks down **maltose** is **maltase**, and the one that breaks down **sucrose** is **sucrase**.

The substance that an enzyme changes is called its **substrate**. The substrate of amylase is starch, and the substrate of lipase is lipids.

Enzyme	Type of enzyme	Substrate	Product(s)
Amylase	Carbohydrase	Starch	Maltose
Maltase	Carbohydrase	Maltose	Glucose
Sucrase	Carbohydrase	Sucrose	Glucose and fructose
Lactase	Carbohydrase	Lactose	Smaller sugars (glucose and galactose)
Protease	Protease	Protein	Amino acids
Lipase	Lipase	Lipids (fats and oils)	Fatty acids and glycerol
Catalase	Not a digestive enzyme; works inside all cells	Hydrogen peroxide	Water and oxygen

Spelling warning: you have to be careful to spell the name of an enzyme and its substrate correctly and to write clearly. Maltose and maltase can look almost the same if you do not write carefully. A useful

rule: names ending in **-ose** are sugars (maltose, sucrose, lactose), and names ending in **-ase** are enzymes (maltase, sucrase, lactase).

How Enzymes Work

Each type of enzyme has molecules with a very specific shape. The enzyme molecule has a 'dent' in it called the **active site**. Figure B4.02 shows how an enzyme catalyses a reaction:

1. The **substrate** molecule has a shape that is **complementary** to the active site of the enzyme (a perfect mirror-image shape).
2. The substrate fits into the active site, where the substrate and the enzyme bind together. For this to happen, the fit has to be perfect.
3. While the substrate is in the active site and bound to the enzyme, the enzyme changes the substrate into a new substance called the **product** (in Figure B4.02, the substrate is split into two product molecules).
4. The product breaks away from the enzyme.
5. The enzyme is now free, and unchanged, so it is ready to bind with another substrate molecule.

In Figure B4.02, one substrate molecule is broken into two product molecules. Enzymes can also catalyse reactions where two substrate molecules bind with the active site and are joined together to form a single product molecule.

Enzymes are very fast. In the few minutes it takes to read about how enzymes work, a single enzyme molecule could have catalysed millions of reactions. Catalase is the fastest enzyme known: one catalase molecule can break down an almost unbelievable 44 million hydrogen peroxide molecules in one second.

The Enzyme–Substrate Complex (Supplement)

The short-lived structure that forms as the substrate slots into the enzyme's active site has its own name: the **enzyme–substrate complex**. The whole process can be summarised as:

- enzyme + substrate → enzyme–substrate complex → enzyme + product(s)

The enzyme appears at both ends of this sequence, which shows that it is unchanged and can be used again.

Activity B4.01: Enzyme Specificity (Supplement)

Each enzyme can only catalyse reactions with one type of substrate. This is called enzyme **specificity**. The activity asks you to think about why enzymes are specific, discuss your ideas with a partner, and be ready to share them with the class.

Q: Why are enzymes specific?

A: The active site of an enzyme has a very specific shape. Only a substrate molecule with a complementary shape can fit into the active site and bind to it to form an enzyme–substrate complex. Molecules of other substances have different shapes, so they do not fit the active site and cannot be changed by that enzyme. This is like a key fitting a lock: only the key with the right shape will fit. For example, amylase breaks down starch, but it cannot break down proteins or fats, because their molecules do not fit its active site.

B4.02 Factors That Affect Enzymes

Optimum Temperature and Optimum pH

Enzymes act very quickly: each enzyme molecule can change many substrate molecules to products every second. But some factors affect this speed of action. Enzymes are very sensitive to **temperature** and **pH**.

Each kind of enzyme has a particular temperature and pH at which it works fastest. These are called the **optimum temperature** and the **optimum pH** for that enzyme. ("Optimum" means "best".)

Figure B4.03a: The Effect of Temperature on Enzyme Activity

Figure B4.03a is a graph of rate of reaction (y-axis, no numbers) against temperature / °C (x-axis, 0 to 60 °C). It shows that:

- At **0 °C** the rate of reaction is zero.
- From 0 °C to about 10 °C, the rate increases only slowly.
- From about 20 °C to about 35 °C, the rate increases steeply.
- The rate is greatest at about **37–38 °C**, just below 40 °C. This peak is labelled **optimum**.
- Above the optimum, the rate falls. It is still about three-quarters of the maximum at 50 °C, then it drops very steeply and reaches **zero at about 58–60 °C**.
- The fall above the optimum is much steeper than the rise below it.

Figure B4.03b: The Effect of pH on Enzyme Activity

Figure B4.03b is a graph of rate of reaction against pH (x-axis, pH 0 to 10). It has two curves:

Feature	A protease in the stomach	Most enzymes
Optimum pH (peak of curve)	About pH 2 (just below 2)	About pH 7
Lowest pH at which it works	About pH 0 (rises steeply from pH 0)	About pH 5.6
Highest pH at which it works	About pH 7 (activity is very low above pH 6)	About pH 10
Shape of curve	Steep rise to the peak, then a gradual fall	Very steep rise from pH 5.6 to 7, then a more gradual fall to pH 10

The graph shows that different enzymes have different optimum pH values. Most enzymes work best in neutral conditions (around pH 7), but the protease in the stomach works best in very acidic conditions.

Buffer Solutions

To investigate the effect of pH, you need to control the pH accurately. A **buffer solution** is a liquid that has a known pH and keeps that pH steady all the time, even if chemical reactions take place. You use a different buffer solution for each pH you want to test.

Experimental Skills B4.01: Investigating the Effect of pH on the Activity of Catalase

Aim

To find out how changing the pH affects the rate at which catalase breaks down hydrogen peroxide to water and oxygen.

How the Experiment Works

- The reaction is: hydrogen peroxide → water + oxygen (catalysed by catalase).
- Catalase is extracted from celery (or any other plant material, such as potato or leaves) by mashing it up in water, so that the catalase forms a solution in the water. It is best if this is done once for the whole class.
- A small square of filter paper is soaked in the catalase solution and pushed to the bottom of a beaker of hydrogen peroxide solution. The catalase on the paper produces oxygen, the oxygen bubbles collect on the paper, and the paper floats upwards (Figure B4.05).

- The faster the catalase works, the sooner the paper rises. So the **time taken for the paper to rise** is a measure of the rate of reaction: a shorter time means a faster rate.
- Buffer solutions are used to set the pH in each beaker.

Apparatus

- A blender or liquidiser, or a pestle and mortar
- Some fresh plant material, such as celery
- Distilled water
- Muslin or a sieve, and a filter funnel
- Some 3% hydrogen peroxide solution
- Five 50 cm^3 beakers
- Small measuring cylinders or syringes, to measure 5 cm^3 and 10 cm^3
- Five buffer solutions, each with a different pH
- 15 small squares of filter paper, all the same size (about 5 mm × 5 mm)
- Forceps (tweezers)
- A stopwatch or timer

Method

Before you start, practise using the forceps to pick up the little pieces of filter paper. It is not easy!

1. Cut up some celery, put it into a blender, add water and blend. (Without a blender, cut the celery into small pieces and grind it with water using a pestle and mortar.)
2. Pour the mixture through muslin in a filter funnel, or through a sieve, to separate most of the bits of celery from the liquid. (Filter paper would take too long.)
3. The liquid now contains catalase. Label a beaker 'catalase solution' and pour some of the liquid into it.
4. Copy the results table.
5. Write the pH values of your buffer solutions into the results table, and label each small beaker with the pH of one of the buffer solutions.
6. Measure 5 cm^3 of hydrogen peroxide solution into each labelled beaker.
7. Measure 10 cm^3 of the appropriate buffer solution into each beaker.
8. Use forceps to pick up a square of filter paper and dip it into the catalase solution (Figure B4.04).
9. Get the stopwatch ready. Still using the forceps, push the paper to the bottom of the hydrogen peroxide solution in one small beaker, let it go, and time how long it takes to rise to the top of the liquid (Figure B4.05). Record the time.
10. Repeat steps 8 and 9 twice more, using the same beaker, so you have three tries.
11. Repeat steps 8, 9 and 10 for each of the other small beakers.
12. Draw a line graph with pH on the x-axis and the mean time for the paper to rise on the y-axis. Join the points with ruled straight lines.

What the figures show: Figure B4.04 shows a filter paper square that has been dipped into the celery extract (catalase solution), held in forceps, ready to go into a beaker of hydrogen peroxide solution and buffer labelled pH 5.6. Figure B4.05 shows the same beaker with a stopwatch running, and oxygen bubbles collecting on the paper and making it float upwards.

Results Table

pH	1st try / s	2nd try / s	3rd try / s	Mean / s
One row for each buffer pH	Time for paper to rise	Time for paper to rise	Time for paper to rise	(1st + 2nd + 3rd) ÷ 3

Variables

Variable	In this experiment
Independent (what you change)	pH, set using five different buffer solutions
Dependent (what you measure)	Time taken for the filter paper to rise to the top of the liquid, in seconds
Controlled (kept the same)	Volume and concentration of hydrogen peroxide (5 cm^3 of 3% solution); volume of buffer (10 cm^3); size of the filter paper squares (about 5 mm × 5 mm); the same catalase solution for every square; temperature (all at room temperature); same size of beaker, so the depth of liquid is the same

Expected Results and Conclusion

- The paper rises fastest (shortest time) at the **optimum pH** of catalase. For catalase from celery or potato, this is usually close to pH 7.
- At pH values further from the optimum, in either direction, the paper takes longer to rise because the catalase works more slowly.
- At a very acidic or very alkaline pH, the paper may not rise at all, because the catalase has been denatured.
- So a graph of mean time against pH is roughly U-shaped, with its lowest point at the optimum pH. This is the opposite way up to a rate graph like Figure B4.03b, because **rate = 1 ÷ time**. For example, if the paper takes 20 s to rise, the rate is 1 ÷ 20 = 0.05 per second.
- **Conclusion:** pH affects the activity of catalase. It works fastest at its optimum pH and more slowly at pH values above or below this.

Safety

- Take care if you are using a blender or liquidiser: keep fingers away from the blades and make sure the lid is on before switching it on.
- Good practice: wear eye protection when using hydrogen peroxide (it is an irritant), and wash any splashes off your skin with water.

Answering the Experiment Questions

Q: (1) Write the word equation for the chemical reaction that took place in the small beakers.

A: hydrogen peroxide → water + oxygen (catalysed by catalase)

Q: (2) Explain, in your own words, why the pieces of filter paper rose to the top of the liquid in the beaker.

A: The filter paper was soaked in catalase. The catalase broke down the hydrogen peroxide in the beaker, producing oxygen gas. Bubbles of oxygen collected on the paper and made it buoyant (less dense than the liquid), so it floated up to the top. The faster the catalase worked, the faster the bubbles formed, and the sooner the paper rose.

Q: (3) Describe the relationship between the pH and the time taken for the filter paper to rise to the top of the liquid.

A: As the pH increases from the lowest value towards the optimum, the time taken for the paper to rise decreases. The shortest time is at the optimum pH (usually around pH 7). As the pH increases above the optimum, the time taken increases again. Quote the actual pH values and mean times from your own results table, with units (s). Remember that the shortest time shows the fastest rate of reaction.

Q: (4) Did you have any anomalous results? If so, explain why you decided that these results were anomalous. Explain how you dealt with these results when you drew your graph.

A: An anomalous result is one that does not fit the pattern: for example, one of the three tries at a pH is very different from the other two, or a mean does not fit the smooth trend of the other points. Possible causes are the paper sticking to the side of the beaker, the paper not being pushed right to the bottom, or more catalase soaking into one square than the others. To deal with it, leave the anomalous result out when calculating the mean (or repeat that try), and plot the mean of the remaining results. For example, if the three tries at one pH were 12 s, 13 s and 31 s, the 31 s result is anomalous, so the mean is (12 + 13) ÷ 2 = 12.5 s. If an anomalous point is plotted on the graph, circle it and do not join it into the line.

Q: (5) Use your results to estimate the optimum pH of the catalase that you used.

A: The optimum pH is the pH with the shortest mean time, which is the lowest point on your graph. For catalase from celery this is usually about pH 7. Because you only tested five pH values, you can only say that the true optimum is close to this value; it may lie between the pH values either side of your lowest point.

Q: (6) Suggest what you could do to obtain a more precise value for the optimum pH of this enzyme.

A: Repeat the experiment using more buffer solutions with pH values closer together around your estimated optimum (for example pH 6.0, 6.5, 7.0, 7.5 and 8.0, or even smaller steps). Do more repeats at each pH and calculate means.

Q: (7) Evaluate your experiment by identifying the main sources of error. Suggest how you could improve the experiment, to reduce these sources of error and give you more confidence that your results are correct.

A: Main sources of error and improvements:

- **Different amounts of catalase on each paper square:** the squares may not be exactly the same size or may be dipped for different times. Cut the squares accurately (or use a hole punch) and dip each one for the same time, for example 5 seconds.
- **Timing by eye:** reaction time when starting and stopping the stopwatch, and deciding exactly when the paper reaches the top. Use the same person and the same end point each time, or video the beakers and time from the recording.
- **The paper may stick to the side** of the beaker or not be released from the very bottom. Release it in the same way, in the centre of the beaker, each time.
- **Hydrogen peroxide is used up:** steps 8 to 10 reuse the same beaker three times, so the hydrogen peroxide becomes slightly less concentrated. Use fresh hydrogen peroxide and buffer for each try.
- **Temperature is not controlled:** stand the beakers in a water-bath at a constant temperature, for example 25 °C.
- **The catalase solution may lose activity over time:** use freshly prepared extract and keep it in the same conditions throughout.
- **Few data points:** test more pH values and do more repeats, then check that the repeats are close together.

Experimental Skills B4.02: Investigating the Effect of Temperature on the Activity of Lipase

Aim and Background

To investigate how temperature affects the activity of lipase. Lipase is an enzyme that catalyses the breakdown of lipids (fats and oils). Milk contains lipids.

- lipid → fatty acids + glycerol (catalysed by lipase)

Fatty acids are acids, so they decrease the pH. Sodium carbonate solution makes the mixture alkaline at the start. The indicator **thymolphthalein** is blue in alkaline conditions and loses its colour when the pH becomes less alkaline. So the **time taken for the blue colour to disappear** shows how quickly the lipase is producing fatty acids: the shorter the time, the faster the lipase is working.

Apparatus

- Fresh milk
- Lipase solution, with a concentration of about 5%
- Sodium carbonate solution, with a concentration of 0.05 mol dm^{-3}
- A small bottle of thymolphthalein indicator, with a dropper
- Five test-tubes and a rack to stand them in
- A test-tube holder
- A syringe or measuring cylinder to measure 10 cm^3
- A small syringe to measure 1 cm^3
- A stopwatch or timer
- One or more glass rods
- Water-baths at five different temperatures, with at least two below the optimum temperature for the enzyme and two above (Figure B4.06 shows an electric water-bath that you can set to a particular temperature, and a simple water-bath made from a beaker of water over a burner)
- A thermometer
- A waterproof marker to label the test-tubes

Method

1. Read through the instructions and draw a results chart that you can fill in as you work.
2. Label your test-tubes with your initials.
3. Measure 5 cm^3 of milk into each test-tube.
4. Measure 7 cm^3 of sodium carbonate solution into each test-tube.
5. Add five drops of thymolphthalein indicator to each test-tube. This makes the liquid go blue. Stir the contents with the glass rod. (Figure B4.07 shows one of these tubes, containing 5 cm^3 of milk, 7 cm^3 of sodium carbonate solution and 5 drops of thymolphthalein indicator, ready to go into the water-bath.)
6. Stand each test-tube in a water-bath at a different temperature. Leave the tubes for a few minutes so that the contents reach the same temperature as the water-bath. Measure this temperature and record it.
7. Using the small syringe, add 1 cm^3 of lipase to the first test-tube. Start timing immediately. Stir with a clean glass rod to mix the lipase into the milk.
8. Record the time taken for the blue colour to disappear, to the nearest second.
9. Repeat for the other four temperatures.
10. Draw a line graph with temperature on the x-axis and time taken on the y-axis. Join the points with ruled lines.

Results Chart

Temperature / °C	Time taken for blue colour to disappear / s
One row for each water-bath temperature	Time to the nearest second

Variables

Variable	In this experiment
Independent (what you change)	Temperature (five water-baths)
Dependent (what you	Time taken for the blue colour to disappear, in seconds

measure)	
Controlled (kept the same)	Volume and type of milk (5 cm^3); volume and concentration of sodium carbonate solution (7 cm^3 of 0.05 mol dm^{-3}); number of drops of indicator (5); volume and concentration of lipase solution (1 cm^3 of about 5%); time allowed for the tubes to reach the water-bath temperature; how the mixture is stirred

Expected Results and Conclusion

- At low temperatures the blue colour takes a long time to disappear.
- As the temperature increases towards the optimum (often somewhere around 35–45 °C for lipase), the time gets shorter.
- Above the optimum, the time gets longer again. At a high temperature (for example 60 °C or more), the colour may never disappear, because the lipase has been denatured.
- So the graph of time against temperature is roughly U-shaped. Its lowest point is at the optimum temperature.
- **Conclusion:** lipase works fastest at its optimum temperature; it works more slowly at lower temperatures, and it is denatured at high temperatures.

Safety

- If you have made your own water-bath, take care not to knock over the beaker of water. Stand up while you are working, so that if something does get knocked over you can quickly move out of the way.
- If the water-bath is very hot, do not touch the hot water, or a hot test-tube, with your hands. Use a test-tube holder.
- Good practice: wear eye protection, because sodium carbonate solution is an irritant.

Reflection

Q: The instructions for these experiments are quite long and complicated. How did you deal with that? What can you do in the future, to help you to understand complicated instructions and follow them successfully?

A: Useful strategies include:

- Read all the instructions through before starting, and highlight the quantities (5 cm^3 milk, 7 cm^3 sodium carbonate, 5 drops of indicator, 1 cm^3 lipase).
- Draw the results chart first (step 1 tells you to), so you know exactly what you are measuring.
- Turn the method into a short flow chart or numbered checklist, and tick off each step as you finish it.
- Set out and label the apparatus in the order you will use it.
- Work with a partner: one reads out each step while the other carries it out.
- Ask your teacher if any step is unclear before you begin, not halfway through.

Answering the Experiment Questions

Q: (1) Explain why the liquid containing the milk was blue at the start of the experiment.

A: The mixture contained sodium carbonate solution, which is alkaline. Thymolphthalein indicator is blue in alkaline conditions, so the liquid was blue.

Q: (2) Explain why the liquid in some of the tubes became colourless.

A: The lipase broke down the lipids (fats) in the milk into fatty acids and glycerol. The fatty acids lowered the pH, so the mixture became less alkaline, and thymolphthalein loses its colour when the pH becomes less alkaline. The colour only disappeared in tubes where the lipase was working. At a temperature where the lipase was denatured (very hot), few or no fatty acids were made, so the liquid stayed blue.

Q: (3) Describe how temperature affected the time taken for the liquid to become colourless.

A: As the temperature increased from the lowest temperature up to the optimum, the time taken for the liquid to become colourless decreased. Above the optimum, the time taken increased again, and at the highest temperature the liquid may not have become colourless at all. Quote your own temperatures and times, with units (°C and s).

Q: (4) According to your results, what is the optimum temperature for lipase?

A: The optimum temperature is the temperature at which the time taken was shortest (the fastest reaction). In most class experiments this is somewhere around 40 °C. The true optimum may lie between two of the temperatures you tested.

Q: (5) Your timer probably measured to 0.1 or even 0.01 seconds. Suggest why it is sensible to record the times to the nearest second.

A: The colour fades gradually, so it is hard to judge the exact moment it disappears, and you also have a reaction time (a fraction of a second) when starting and stopping the timer. The times are therefore not accurate to 0.1 s or 0.01 s. Recording those extra digits would suggest a precision that the measurements do not really have, so the nearest second is a sensible level of precision.

Q: (6) Evaluate your experiment, by identifying the main sources of error. How could you improve the experiment, so that you have more confidence in your results?

A: Main sources of error and improvements:

- **Judging the end point:** it is difficult to decide exactly when the blue colour has gone. Compare each tube with a colourless reference tube against a white background, or use a pH meter or colorimeter.
- **Lipase added at room temperature:** this cools or warms the mixture. Stand the lipase solution in the same water-bath before adding it.
- **Water-bath temperature changes:** a beaker over a burner is hard to keep steady. Use a thermostatically controlled electric water-bath, and check the temperature with a thermometer throughout.
- **Delay between adding lipase and starting the timer, and uneven stirring:** start the timer as the lipase goes in, and stir every tube in the same way.
- **Only one result at each temperature:** do three repeats at each temperature, calculate means, and test more temperatures, especially close to the optimum.

Self Assessment

The book asks you to rate how well you constructed your results chart, using a smiling face (did it really well), a straight face (made a good attempt and partly succeeded) or a sad face (did not try or did not succeed), for each of these statements:

- I drew the results chart using a ruler for all the rows and columns.
- I included the units (s for the time measurements, and °C for the temperature measurements) in the headings of the table.
- I did not write the units again with the individual results in the table.
- I recorded each temperature reading to the nearest whole degree or 0.5 degree.
- I recorded each time reading to the nearest second.
- A friend confirms that my results chart is very easy to understand.

Denaturation

Temperature. Enzymes have an optimum temperature at which they work fastest:

- Enzymes from the human body generally have an optimum temperature of about **37 °C**.
- Enzymes from plants may have optimum temperatures much lower than this.
- Some bacteria, especially those that live in hot springs, have very high optimum temperatures: up to **80 °C** in some cases.

For many enzymes, a temperature above about **60 °C** completely stops them working. This is because the high temperature damages the enzyme. The enzyme is said to be **denatured**, and it cannot catalyse its reaction any more.

pH. Enzymes also have an optimum pH. For most enzymes this is around **pH 7**, but some enzymes work best at a much higher or lower pH. For example, in Chapter B6 you will meet an enzyme that works best in the acidic conditions of the human stomach, at a pH of about **2**. When an enzyme is placed in a liquid with a pH that is not its optimum pH, it is damaged. The further the pH is from the optimum, the more the enzyme is damaged, and well away from the optimum it is denatured and cannot catalyse its reaction.

Important: low temperatures do NOT denature enzymes. At 0 °C an enzyme has almost no activity because its molecules are moving very slowly, but if it is warmed up again it will start working. Denaturation is caused by high temperatures or by a pH far from the optimum.

Explaining the Effect of Temperature on Enzyme Activity (Supplement)

Use Figure B4.03a to follow these steps:

1. **At 0 °C, there is no activity.** Molecules have very little kinetic energy and move only slowly. Substrate molecules rarely collide with the enzyme, so they rarely enter its active site, and very few substrate molecules are converted to product.
2. **As temperature increases, activity increases.** The kinetic energy of the enzyme and substrate molecules increases. They move faster and collide with each other more frequently and with more energy, so effective collisions are more frequent. Each second, more substrate molecules collide with an active site, form enzyme–substrate complexes and are converted to product.
3. **At the optimum (about 37–38 °C in Figure B4.03a), activity is greatest.** There are the most effective collisions per second while the active site still has its correct shape.
4. **Above the optimum, activity decreases.** The kinetic energy of the enzyme begins to shake it apart. Its molecules begin to lose their shape, so the active site is no longer a true complementary shape to the substrate. When a substrate molecule collides with an active site it may not fit, so it cannot form an enzyme–substrate complex and is not converted to product.
5. **At about 60 °C, activity stops.** The active site is so out of shape that the enzyme has completely stopped working. It is completely denatured.

Explaining the Effect of pH on Enzyme Activity (Supplement)

Something similar happens when pH changes, but the explanation is about **shape**, not kinetic energy:

- Each enzyme molecule has a very specific shape, and its active site is the best fit for the substrate at the enzyme's optimum pH.
- As the pH moves away from the optimum, the enzyme's shape changes, so the active site fits the substrate less well and fewer enzyme–substrate complexes form.
- A pH well above or below the optimum causes the enzyme molecule to lose its shape completely (denature), so it can no longer bind with the substrate.

Comparing the Effects of Temperature and pH

Condition	Effect of temperature	Effect of pH
Below the optimum	Molecules have less kinetic energy, so there are fewer effective collisions; the enzyme is slow but NOT denatured	More acidic than the optimum: the enzyme's shape changes, so the active site fits the substrate less well; far below the optimum the enzyme is denatured
At the optimum	Fastest rate: most effective collisions while the active site still has its correct shape	Fastest rate: the active site is the best fit for the substrate

Above the optimum	The enzyme starts to lose its shape; the active site is no longer complementary; it is completely denatured at about 60 °C for many enzymes	More alkaline than the optimum: the enzyme's shape changes, so the active site fits the substrate less well; far above the optimum the enzyme is denatured

Project B4.01: An Enzyme Demonstration

Working in a group of three, you plan a short demonstration to explain what enzymes are and how they work. It can be 'live' with a commentary, or a video with a voiceover. Imagine that your audience knows nothing about enzymes: your task is to get them interested and help them begin to understand what enzymes are and what they do.

Choosing an enzyme: the best enzyme to use is **catalase**, because it is easy to see it working. Catalase is present in all animal and plant tissues, and when it breaks down hydrogen peroxide it releases oxygen.

- **Figure B4.08:** raw liver contains a lot of catalase, so it breaks down hydrogen peroxide and releases oxygen very quickly. The reaction can be quite dramatic, with froth pouring out of the flask.
- **Figure B4.09:** if hydrogen peroxide is added to raw potato, the catalase in the potato breaks it down and releases oxygen, which forms a white froth. This is a quieter reaction, which may be better for a demonstration.

Ideas for the demonstration: show the froth forming on raw potato; explain the active site with a simple model (for example, two shapes that fit together like a lock and key); compare raw potato with boiled potato, where the catalase has been denatured and no froth forms; show that the gas relights a glowing splint, proving it is oxygen.

Q: It is extremely important to consider safety as you plan. What are the risks that you need to be aware of? How will you reduce these risks?

A: Risks and how to reduce them:

- **Hydrogen peroxide** is an irritant and can harm the eyes and skin. Wear eye protection and gloves, use only dilute (3%) solution, and wipe up spills straight away.
- **Froth overflowing**, especially with liver. Use small amounts in a large container standing on a tray, and never put a lid on the container (the oxygen produced would build up pressure).
- **Raw liver** may carry harmful bacteria. Wear gloves, wash your hands afterwards, and dispose of the liver safely.
- **Knives** for cutting potato or liver. Cut on a board, cutting away from your body.
- **Boiling water** (if you boil potato for comparison). Handle hot liquids carefully and let them cool.
- **Glowing splints** are a fire risk. Tie back long hair and keep flammable materials away.

Key Formulas and Concepts

1. **Catalyst:** a substance that increases the rate of a chemical reaction and is not changed by the reaction; enzymes are biological catalysts made of protein.
2. **How an enzyme works:** enzyme + substrate → enzyme–substrate complex → enzyme + product(s)
3. **Amylase:** starch → maltose
4. **Maltase:** maltose → glucose
5. **Protease:** protein → amino acids
6. **Lipase:** lipid (fats and oils) → fatty acids + glycerol
7. **Catalase:** hydrogen peroxide → water + oxygen
8. **Lactase:** lactose → smaller sugars (glucose + galactose), which are absorbed into the blood
9. **Rate and time:** rate = 1 ÷ time taken. A shorter time means a faster rate. Example: a time of 4 min gives a rate of 1 ÷ 4 = 0.25 per minute, and a time of 8 min gives 1 ÷ 8 = 0.125 per minute, so the

first reaction is twice as fast.

10. **Optimum temperature:** about 37 °C for human enzymes; many enzymes are denatured above about 60 °C; some bacteria in hot springs have optimum temperatures up to 80 °C.
11. **Optimum pH:** about pH 7 for most enzymes; about pH 2 for the protease in the stomach.
12. **Range:** the lowest to the highest value; for example, most enzymes in Figure B4.03b work from about pH 5.6 to about pH 10.
13. **Naming rule:** enzyme names usually end in -ase (carbohydrase, protease, lipase, amylase); sugar names end in -ose (maltose, sucrose, lactose).
14. **Iodine test for starch:** blue-black means starch is present; brown (the colour of iodine solution itself) means no starch is present.
15. **Thymolphthalein:** blue in alkaline conditions; it loses its colour when the pH becomes less alkaline (for example, when fatty acids are made).

Exam Focus Priorities

1. **Learn the definition of a catalyst exactly** — "a substance that increases the rate of a chemical reaction and is not changed by the reaction". You need both parts: it speeds up the reaction and it is unchanged.
2. **Enzymes are proteins** — they are made of amino acids and act as biological catalysts in all metabolic reactions. Without them, reactions would be too slow to sustain life.
3. **Use the word complementary** — the active site has a shape that is complementary to the substrate. Do not say the active site is "the same shape" as the substrate.
4. **Name the enzyme–substrate complex** — in Supplement answers, say that the substrate binds to the active site to form an enzyme–substrate complex, the products are released, and the enzyme is unchanged.
5. **Explain specificity using shape** — only a substrate with a shape complementary to the active site can fit and bind, so each enzyme acts on only one type of substrate.
6. **Enzymes are denatured, not killed** — enzymes are molecules, not living things, so never write that they "die" or are "killed".
7. **Low temperature does not denature enzymes** — at low temperatures enzymes are slow because molecules have little kinetic energy; they work again when warmed. Denaturation happens above the optimum temperature or at a pH far from the optimum.
8. **Use the key words for temperature** — kinetic energy, frequency of effective collisions, enzyme–substrate complex, shape of the active site, complementary, denatured.
9. **Explain pH effects in terms of shape** — pH changes the shape of the enzyme and its active site. Kinetic energy and collisions are the explanation for temperature, not for pH.
10. **Describe graphs with numbers and units** — quote the optimum (for example "about 37 °C") and the values where activity starts and stops. "Describe" means say what happens; "explain" means give the reasons.
11. **Time and rate are opposites** — a shorter time means a faster rate (rate = 1 ÷ time). In the catalase, lipase and amylase experiments, the shortest time is at the optimum.
12. **Know the iodine test colours** — blue-black means starch is present; brown means no starch. Do not say that iodine "turns clear".
13. **Spell enzyme names carefully** — maltose (a sugar) and maltase (an enzyme) look almost the same; lipid is the substrate and lipase is the enzyme.
14. **Make controlled variables specific** — write "temperature", "volume of amylase solution" or "concentration of starch solution", not just "the amount".
15. **Different enzymes have different optimums** — most enzymes about pH 7 but the stomach protease about pH 2; human enzymes about 37 °C but some hot-spring bacteria up to 80 °C.

Textbook Questions with Answers

Question B4.01

Explain why enzymes are called biological catalysts.

Answer: Enzymes are catalysts because they increase the rate of chemical reactions without being changed by the reactions, so each enzyme molecule can be used again and again. They are called biological catalysts because they are proteins made by living cells, and they catalyse the metabolic reactions that take place inside living organisms. Without enzymes, these reactions would happen very slowly or not at all, far too slowly to sustain life.

Question B4.02

You read about the enzyme lactase at the start of this chapter.

a) What type of enzyme is lactase?

b) Name the substrate of lactase.

Answer:

a) Lactase is a carbohydrase, because it breaks down a carbohydrate (the sugar lactose) into smaller sugar molecules.

b) Lactose, the sugar found in milk.

Question B4.03

Look at Figure B4.03a. Describe what is shown on the graph. (When you are describing a graph, it is a good idea to quote figures, with units.)

(Figure B4.03a is a graph of rate of reaction, with no scale, against temperature / °C from 0 to 60 °C. A single curve rises to a peak, which is labelled 'optimum', and then falls.)

Answer: The graph shows how the rate of an enzyme-controlled reaction changes with temperature:

- At 0 °C the rate of reaction is zero.
- As the temperature increases from 0 °C, the rate increases, slowly at first (up to about 10 °C) and then steeply between about 20 °C and 35 °C.
- The rate reaches a maximum at about 37–38 °C (just below 40 °C). This is the optimum temperature.
- Above the optimum, the rate decreases, gradually at first and then very steeply, until it reaches zero at about 58–60 °C.
- The rate falls much more steeply above the optimum than it rises below it.

Question B4.04

Look at Figure B4.03b.

a) What is the optimum pH of the protease in the stomach?

b) What is the optimum pH of most enzymes?

c) What is the range of pH values at which enzymes are able to work?

(Figure B4.03b is a graph of rate of reaction against pH, from pH 0 to pH 10. It has two curves: one for a protease in the stomach, and one labelled 'most enzymes'.)

Answer:

a) About pH 2. The peak of the protease curve is just below pH 2.

b) About pH 7.

c) For most enzymes, the range is from about pH 5.6 to about pH 10; below about pH 5.6 and above about pH 10 their activity is zero. The protease in the stomach has a different range, from about pH 0 to

about pH 7. So, looking at both curves, enzymes can work across almost the whole range shown (about pH 0 to pH 10), but each enzyme works only over its own narrower range.

Question B4.05

Look at the graph that you drew from your results in Experimental skills B4.02 Investigating the effect of temperature on the activity of lipase. In your own words, explain the shape of the graph (use your biological knowledge and understanding to give reasons for the shape of the graph).

Answer: The graph of time taken for the blue colour to disappear (y-axis) against temperature (x-axis) should be roughly U-shaped. The time is long at low temperatures, falls to a minimum at the optimum temperature (often around 40 °C), and then rises again at higher temperatures. At the highest temperature the colour may not have disappeared at all. A short time means a fast rate, so this graph is the upside-down version of a rate graph like Figure B4.03a.

- **Low temperatures (long times):** lipase and lipid molecules have little kinetic energy and move slowly, so they rarely collide. Few lipid molecules enter the active site each second, so fatty acids are made slowly and it takes a long time for the pH to fall enough to make the thymolphthalein colourless.
- **Rising towards the optimum (times get shorter):** the molecules gain kinetic energy, move faster, and collide more often and with more energy. There are more effective collisions, so more enzyme–substrate complexes form and more fatty acids are made per second. The pH falls faster, so the colour disappears sooner.
- **At the optimum (shortest time):** the rate is greatest, because there are the most effective collisions while the active site still has its correct shape.
- **Above the optimum (times get longer):** the high kinetic energy starts to shake the lipase molecules apart. They lose their shape, so the active site is no longer complementary to the lipid molecules. Fewer substrate molecules fit and form enzyme–substrate complexes, so fatty acids are made more slowly.
- **Very high temperatures (no colour change):** the lipase is completely denatured, so no fatty acids are made, the pH does not fall and the liquid stays blue.

Question B4.06

Use your own words to explain the shapes of the curves for most enzymes, in Figure B4.03b.

(In Figure B4.03b, the curve for most enzymes is at zero below about pH 5.6, rises steeply to a peak at about pH 7, and then falls back to zero at about pH 10.)

Answer:

- **At the optimum pH (about pH 7):** the enzyme molecule has exactly the right shape, so its active site is the best fit for (complementary to) the substrate. Substrate molecules fit easily, many enzyme–substrate complexes form, and the rate of reaction is highest.
- **As the pH moves away from 7 (more acidic or more alkaline):** the change in pH begins to change the shape of the enzyme molecule, including its active site. The active site becomes less complementary to the substrate, so fewer substrate molecules can fit and bind, fewer enzyme–substrate complexes form, and the rate falls.
- **Well below or well above the optimum (below about pH 5.6 or above about pH 10):** the enzyme has lost its shape completely; it is denatured. The substrate can no longer bind with the active site, so the rate is zero.
- The curve rises more steeply on the acidic side (pH 5.6 to 7) than it falls on the alkaline side (pH 7 to 10). This shows that these enzymes are affected more by a small decrease in pH below 7 than by a similar increase above 7.

Practice Exam Questions with Answers

Practice Question 1

Identify the smaller molecules from which an enzyme molecule is made. (1 mark)

A) amino acids

B) fatty acids

C) glucose

D) glycerol

Answer: A — amino acids. Enzymes are proteins, and protein molecules are made from long chains of amino acids [1]. Fatty acids and glycerol are the smaller molecules that make up lipids, and glucose is the smaller molecule that makes up starch.

Practice Question 2

(5 marks)

a) Complete the sentences about enzymes. Choose words from the list. [3]

Word list: chemical, proteins, metabolic, active, complementary

Enzymes are _____ that function as biological catalysts.

Enzymes are involved in all _____ reactions.

The part of an enzyme where a substrate binds is called its _____ site.

b) Describe why enzymes are important to all living organisms. [2]

Answer:

a) Enzymes are **proteins** that function as biological catalysts [1]. Enzymes are involved in all **metabolic** reactions [1]. The part of an enzyme where a substrate binds is called its **active** site [1]. ("Chemical" is not accepted for the second gap, because enzymes are not involved in all chemical reactions, only in metabolic reactions inside living things.)

b) Any two of these points:

- Enzymes speed up (catalyse) the metabolic reactions in living organisms [1].
- Without enzymes, these reactions would happen too slowly or not at all; enzymes make the reactions fast enough to sustain life [1].
- They allow essential processes to happen quickly enough, for example digesting food so that it can be absorbed, building proteins from amino acids, or breaking down toxic hydrogen peroxide (catalase) [1].

Practice Question 3

The figure below shows some stages in an enzyme-controlled reaction. (4 marks)

(The figure shows three stages, from top to bottom. Stage 1: an enzyme molecule (labelled 'enzyme') with a dent in one side, labelled Q, and a separate round molecule labelled P. Stage 2: the round molecule fitted into the dent of the enzyme, with the two together labelled R. Stage 3: the enzyme, with its dent empty again, next to the round molecule now split into two pieces, labelled S.)

Identify the parts labelled P, Q, R and S. [4]

P: _____

Q: _____

R: _____

S: _____

Answer:

P = substrate [1]

Q = active site [1]

R = enzyme–substrate complex [1]

S = products (the product molecules) [1]

Practice Question 4

Amylase is an enzyme that breaks down starch to maltose. (9 marks)

a) A student put some starch solution into a test-tube. He took a small sample of starch from the tube and added it to a drop of iodine solution on a white tile.

(The figure shows a test-tube of starch solution, a glass rod carrying a sample of starch solution, and a white tile with drops of iodine solution in its dimples.)

Next, he added some amylase solution to the starch solution. After five minutes, he took another sample from the tube, and tested it with iodine solution.

The table shows his results.

	Result of iodine test
At start	blue-black
After adding amylase solution	brown

Explain the student's results. [3]

b) The student investigated the effect of pH on the activity of amylase.

He placed the same volume of starch solution in five tubes. He added different buffer solutions to each tube. He then added amylase solution to each tube.

He took samples of the mixture from each tube and measured the time for all the starch to disappear.

His results are shown in the table.

pH	Time for starch to disappear / min
4	did not disappear
6	8
7	4
9	14
11	did not disappear

i) Use the student's results to describe the effect of pH on the activity of amylase. [3]

ii) The student kept the volume of starch solution the same in each tube. State three other variables that the student should have kept the same in his experiment. [3]

Answer:

a) Any three of these points:

- At the start, the sample turned the iodine solution blue-black, which shows that starch was present [1].
- Amylase broke down (digested) the starch [1] into maltose [1].
- After five minutes the iodine solution stayed brown, which shows that there was no starch left; maltose does not give a blue-black colour with iodine [1].

b) i) One mark for each of these points:

- Amylase is most active at pH 7, which is its optimum pH; the starch disappeared in the shortest time (4 min) [1].

- Activity decreases as the pH moves away from 7 in either direction: the starch took 8 min to disappear at pH 6 and 14 min at pH 9 [1].
- At pH 4 and pH 11 the starch did not disappear, so the amylase was not working (it was denatured) at these pH values [1].
- Working with rates: rate = 1 ÷ time, so the rate was 1 ÷ 8 = 0.125 per min at pH 6, 1 ÷ 4 = 0.25 per min at pH 7, and 1 ÷ 14 = 0.071 per min at pH 9. Amylase worked twice as fast at pH 7 as at pH 6 (8 ÷ 4 = 2), and 3.5 times as fast as at pH 9 (14 ÷ 4 = 3.5).

b) ii) Any three of these variables [1 each]:

- temperature (for example, keep all the tubes in the same water-bath)
- volume of amylase solution
- concentration of amylase solution
- concentration of starch solution
- volume of buffer solution
- how often samples were taken and tested (and the size of each sample)

Practice Question 5

The graph below shows how the activity of two enzymes, A and B, is affected by temperature. (12 marks)

(The graph plots rate of activity, with no scale, on the y-axis against temperature / °C on the x-axis, from 0 to 100 °C. Enzyme A: activity rises from zero at 10 °C to a peak at 40 °C, then falls steeply to zero at about 58 °C. Enzyme B: activity rises from zero at 40 °C to a peak at 70 °C, then falls to zero at about 92 °C. The two peaks are at about the same height, and the two curves cross at about 55 °C.)

a) Compare the effect of temperature on enzyme A and enzyme B. [5]

b) Explain the effect of temperature on enzyme B from:

i) 40 to 60 °C [3]

ii) 70 to 90 °C. [4]

Answer:

a) Any five of these points (a comparison needs similarities and differences, and each statement should mention both enzymes):

- Similar pattern: for both enzymes, activity increases as temperature increases up to an optimum, and then decreases [1].
- Enzyme A has an optimum temperature of 40 °C, whereas enzyme B has an optimum of 70 °C, which is 30 °C higher [1].
- Enzyme A starts to work at 10 °C, but enzyme B shows no activity until 40 °C [1].
- Enzyme A stops working (is completely denatured) at about 58 °C, whereas enzyme B keeps working until about 92 °C [1].
- Both enzymes reach about the same maximum rate of activity [1].
- Both work over a similar range of about 50 °C: A from 10 °C to about 58 °C, and B from 40 °C to about 92 °C [1].
- Below about 55 °C enzyme A is more active than enzyme B, and above about 55 °C enzyme B is more active; they have the same activity at about 55 °C [1].
- For both enzymes, activity falls more steeply above the optimum than it rises below it [1].

b) i) Three of these points:

- The activity of enzyme B increases, from zero at 40 °C to about 70% of its maximum at 60 °C [1].
- As temperature increases, the enzyme and substrate molecules gain kinetic energy and move faster [1].

- They collide more frequently, so there are more effective collisions between substrate molecules and the active site [1].
- So more enzyme–substrate complexes form and more product is made per second [1]. (This range is below B's optimum, so B is not being denatured.)

b) ii) Four of these points:

- The activity of enzyme B decreases, falling from its maximum at 70 °C to a low level by 90 °C (and to zero at about 92 °C) [1].
- Above the optimum, the high kinetic energy makes the enzyme molecule vibrate so much that it begins to shake apart [1].
- The enzyme, including its active site, changes shape [1].
- The active site is no longer complementary to the substrate, so the substrate cannot fit or bind [1].
- Fewer enzyme–substrate complexes form, so less product is made per second [1].
- The enzyme becomes denatured; by about 92 °C it is completely denatured and has no activity [1].

Summary of Key Concepts

Chapter Summary

- **Enzymes are proteins** that function as biological catalysts and increase the rate of metabolic reactions.
- **Without enzymes**, these reactions would not take place fast enough to sustain life.
- **Each enzyme has an active site** that has a complementary shape to its substrate.
- **When the substrate binds with the active site** of the enzyme, the substrate is converted to product. The enzyme is unchanged.
- **The binding of substrate to the active site** of the enzyme forms an enzyme–substrate complex.
- **Enzymes have an optimum temperature and pH** at which they work most quickly. Higher temperatures or a very different pH cause the enzyme to change shape and denature.
- **As temperature increases**, the kinetic energy of enzyme and substrate molecules increases, and they collide more frequently. This increases the rate of activity of the enzyme. However, above the optimum temperature, the enzyme begins to lose its shape, until it can no longer combine with the substrate. The shape of the active site is no longer complementary to the shape of the substrate.

Command Words Used in This Chapter

- **Identify** — name, select or recognise.
- **Describe** — state the points of a topic; give characteristics and main features.
- **Explain** — set out purposes or reasons; make the relationships between things clear; say why and/or how, and support with relevant evidence.
- **State** — express in clear terms.

Exam Technique Tips

- **Graph questions:** "describe" means say what the graph shows, with numbers and units (for example, "the activity of enzyme A increases from zero at 10 °C to a maximum at 40 °C, then falls to zero at about 58 °C"). "Explain" means give the biological reasons (kinetic energy, collisions, shape of the active site, denaturation).
- **Compare questions:** write about both enzymes in every sentence, using words like "whereas" and "both", and include at least one similarity and one difference.
- **Time versus rate:** if a table gives times, remember that the shortest time shows the highest activity.
- **Experiment questions:** name specific controlled variables, give a reason for any anomalous result, and suggest realistic improvements (repeats, smaller intervals, a water-bath, fresh solutions).

- **Diagrams of enzyme action:** label the substrate, active site, enzyme–substrate complex and products, and say that the enzyme is unchanged at the end.

Self-Evaluation Checklist

Check that you can:

- explain what a catalyst is (B4.01)
- explain why enzymes are said to be biological catalysts, and why they are important to sustain life (B4.01)
- describe how enzymes work, using the terms active site, substrate, complementary shape and product (B4.01)
- use the term enzyme–substrate complex (B4.01)
- explain why enzymes are specific (B4.01)
- describe how to investigate the effect of temperature and pH on enzyme activity, and describe these effects, using the words optimum and denaturation (B4.02)
- explain the effect of temperature on enzyme activity, using the words kinetic energy, frequency of effective collision, complementary shape and denaturation (B4.02)
- explain the effect of pH on enzyme activity, using the words complementary shape and denaturation (B4.02)